CHAPTER 13

NUCLEI

CHAPTER ESSENCE AT A GLANCE:
Nuclear composition, size, mass defect, binding energy per nucleon, radioactivity laws, and nuclear fission and fusion.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Nuclear Volume & Constant Matter Density:** Nuclear radius scales with the cube root of mass number, meaning nuclear matter density remains identical for all atomic nuclei.

• **Strong Nuclear Force:** Short-range attractive nuclear force binds protons and neutrons together, exhibiting charge independence and overwhelming strength over electrostatic repulsion.

• **Mass Defect & Binding Energy:** Total rest mass of a stable nucleus is always less than the sum of its individual nucleons; this mass defect accounts for nuclear binding energy.

• **Binding Energy Curve Significance:** Peaks around iron-56, explaining why heavy unstable nuclei release enormous energy by fissioning, while light nuclei release energy by fusing together.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$R = R_0 A^{1/3} \quad (R_0 \approx 1.2 \text{ fm})$$

Nuclear radius as a function of mass number A

$$E = \Delta m \cdot c^2 = \Delta m \times 931.5 \text{ MeV}$$

Mass-energy equivalence and nuclear binding energy

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{\text{nucleus}}$$

Nuclear mass defect formula

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** To prove density is constant: Mass M = A · m; Volume V = (4 divided by 3)π R_0^3 A; Density ρ = M divided by V = m divided by [(4 divided by 3)π R_0^3] (A cancels out!).

• **Alert:** [Valuation Rubric] Draw Binding Energy per nucleon curve showing peak at Fe-56 (8.75 MeV) .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Composition of nucleus, atomic mass unit (amu or u), and mass of proton/neutron [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Nuclear Composition]** A nucleus contains protons and neutrons, collectively termed nucleons. Atomic number Z = number of protons; Mass number A = total nucleons (Z + N); Neutron number N = A - Z. [1 Mark]

• **[Atomic Mass Unit (u or amu)]** Defined as exactly 1 divided by 12th of the mass of an unbound carbon-12 (^{12}C) atom at rest: 1 u = (12 × 10^{-3} kg/mol) divided by (12 × 6.022 × 10^{23} mol^{-1}) = 1.660539 × 10^{-27} kg. Proton mass: m p = 1.007276 u = 1.6726 × 10^{-27} kg. Neutron mass: m n = 1.008665 u = 1.6749 × 10^{-27} kg. [1 Mark]

• **Key Exam Result:** 1 u = 1.6605 × 10^{-27} kg; Nucleus consists of Z protons and (A - Z) neutrons

2. Nuclear radius formula R = R 0 A$^{\text{one-third}}$ and proof that nuclear density is constant independent of A [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Nuclear Radius Formula]** From α-scattering experiments, nuclear radius is: R = R_0 A^(one-third), where $R_0 \approx 1.2 \times 10^{-15}$ m = 1.2 fm and A is mass number. [1 Mark]

• **[Proof that Nuclear Density is Constant]** Nuclear volume: V = (4 divided by 3) π R^3 = (4 divided by 3) π (R_0 A^(one-third))^3 = (4 divided by 3) π R_0^3 A. Mass of nucleus: M = A · m avg (where m avg ≈ 1.66 × 10^{-27} kg). Nuclear density: ρ = Mass or Volume = (A · m avg) divided by [(4 divided by 3) π R_0^3 A] = m avg divided by [(4 divided by 3) π R_0^3]. Notice that mass number A completely cancels out! ρ = (1.66 × 10^{-27}) divided by [(4 divided by 3) × 3.1416 × (1.2 × 10^{-15})^3] = 2.3 × 10^{17} kg/m^3. Nuclear density is CONSTANT for all nuclei, independent of mass number A! [2 Marks]

• **Key Exam Result:** Nuclear density ρ ≈ 2.3 × 10^{17} kg/m^3 (constant for all elements)

3. Einstein's mass-energy equivalence E = mc^2 and energy equivalent of 1 amu = 931.5 MeV [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Mass-Energy Equivalence]** Einstein's special theory of relativity states mass can be converted to energy and vice-versa: E = m c^2. [1 Mark]

• **[Energy Equivalent of 1 amu in MeV]** m = 1 u = 1.660539 × 10^{-27} kg, c = 2.9979 × 10^8 m/s. E = m c^2 = (1.660539 × 10^{-27} kg) × (2.9979 × 10^8 m/s)^2 = 1.4924 × 10^{-10} Joules. Converting to MeV (1 eV = 1.6022 × 10^{-19} J, 1 MeV = 1.6022 × 10^{-13} J): E = (1.4924 × 10^{-10} J) divided by (1.6022 × 10^{-13} J/MeV) = 931.5 MeV. [1 Mark]

• **Key Exam Result:** 1 amu (u) is equivalent to 931.5 MeV of energy

4. Einstein's mass-energy equivalence E = mc^2 and energy equivalent of 1 amu = 931.5 MeV [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Mass-Energy Equivalence]** Einstein's special theory of relativity states mass can be converted to energy and vice-versa: E = m c^2. [1 Mark]

• **[Energy Equivalent of 1 amu in MeV]** m = 1 u = 1.660539 × 10^{-27} kg, c = 2.9979 × 10^8 m/s. E = m c^2 = (1.660539 × 10^{-27} kg) × (2.9979 × 10^8 m/s)^2 = 1.4924 × 10^{-10} Joules. Converting to MeV (1 eV = 1.6022 × 10^{-19} J, 1 MeV = 1.6022 × 10^{-13} J): E = (1.4924 × 10^{-10} J) divided by (1.6022 × 10^{-13} J/MeV) = 931.5 MeV. [1 Mark]

• **Key Exam Result:** 1 amu (u) is equivalent to 931.5 MeV of energy

5. Mass defect \Δm and Nuclear Binding Energy E b = \Δm c^2 [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Mass Defect (Δm)]** The difference between the sum of the rest masses of individual constituent nucleons and the actual rest mass of the bound nucleus: Δm = [Z m p + (A - Z) m n] - M nucleus. (Always positive: Δm > 0). [1.5 Marks]

• **[Nuclear Binding Energy (E b)]** The energy equivalent of the mass defect: E b = Δm · c^2 = [Z m p + (A - Z) m n - M] c^2 = Δm (in u) × 931.5 MeV. Binding energy per nucleon: E bn = E b divided by A. Measures the intrinsic stability of a nucleus against disassembly. [1.5 Marks]

• **Key Exam Result:** Δm = [Z m p + (A-Z) m n] - M; E b = Δm × 931.5 MeV; E bn = E b divided by A

6. Nuclear forces: key properties (short range, strong, saturated, charge independent, non-central) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Five Key Properties of Nuclear Forces]** 1. Strongest Force in Nature: Roughly 100 times stronger than electrostatic Coulomb repulsion and 10^{38} times stronger than gravitational force. 2. Short-Range Force: Operates strictly over subatomic nuclear distances ($r \leq 1$ - 2 fm). At $r > 3$ fm, it vanishes to zero. Below 0.8 fm, it becomes strongly repulsive. 3. Charge Independent: Attractive force between proton-proton (p-p), neutron-neutron (n-n), and proton-neutron (p-n) is identical. 4. Non-Central and Spin-Dependent: Does not act solely along the line joining nucleon centers; depends on nucleon intrinsic spins. 5. Saturation Property: A nucleon interacts only with its immediate nearest neighbors, keeping nuclear density constant. [3 Marks]

• **Key Exam Result:** Strongest, short-range (<2 fm), charge-independent, saturated, non-central

7. Numerical on calculation of mass defect and binding energy per nucleon of iron/helium [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula]** Binding energy $E_b = [Z m_p + (A - Z) m_n - M] \times 931.5$ MeV. [1 Mark]

• **[Step 2: Calculation for Helium-4]** Constituents: $Z = 2$ protons, $(A - Z) = 2$ neutrons. Mass of 2 protons = 2×1.007276 u = 2.014552 u. Mass of 2 neutrons = 2×1.008665 u = 2.017330 u. Sum of nucleon masses = 4.031882 u. Actual mass of ${}^{4}_{2}He$ nucleus = 4.001506 u. Mass defect $\Delta m = 4.031882 - 4.001506 = 0.030376$ u. Total Binding Energy $E_b = 0.030376 \times 931.5$ MeV = 28.30 MeV. Binding energy per nucleon: $E_{bn} = 28.30$ divided by 4 = 7.07 MeV/nucleon. [2 Marks]

• **Key Exam Result:** $\Delta m = 0.03038$ u; $E_b = 28.3$ MeV; $E_{bn} = 7.07$ MeV/nucleon